

02 · Does the offer work better for some segments? — segment effects (pathmc)

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The business decision. The email lifts spend *on average* (that’s the ATE from notebook 01). But an average can hide everything that matters: maybe the email really moves **high-value, highly-engaged** customers and barely touches the rest. If the segments genuinely differ, we should write segment-specific send rules; if the apparent “gap” is just noise, chasing it would be overfitting our marketing to randomness. Either way we want to know **what it costs to treat everyone as the average**.

1.0.1 The concept: moderation and the CATE

When a treatment’s effect *depends on who receives it*, we say the effect is **moderated** by the customer’s characteristics, and the characteristic (here, being high-value, or an engagement score) is a **moderator**. Formally we’re estimating the **CATE** — the *Conditional Average Treatment Effect*, $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$: the average email effect *among customers who look like x* . ($Y(1), Y(0)$ are the two potential outcomes — spend if emailed / not — from notebook 01; $\mathbb{E}[\cdot]$ is “average over customers.”) The ATE is just the CATE averaged over everybody.

1.0.2 The tool: a structural model + the do-operator

Notebook 01 used flexible trees (BART) to get the CATE. Here we use a different, more *interpretable* approach — a **structural causal model** via the `pathmc` library. We write down the equation we believe generated spend (including an **interaction term** — a feature that multiplies the treatment, which is exactly how “the effect depends on a moderator” is encoded), fit it, and then ask the model counterfactual questions with the **do-operator**: `do(email=1)` means “*intervene* to email this customer,” as opposed to merely *observing* emailed customers (who differ in other ways). The difference in predicted spend between `do(email=1)` and `do(email=0)`, at a given x , *is* the CATE at x .

The single most important output is the **posterior of the interaction coefficient**: if it’s clearly away from zero, the segments *really* respond differently; if it straddles zero, we can’t tell the gap from noise — so we shouldn’t build segment rules on it (and might instead run a bigger test). That’s the “is the difference real?” test, answered with uncertainty.

7-step contract: question -> simulate -> identify -> estimate -> validate -> decide in € -> caveats.

1.1 2 · Simulate a ground truth

Customers are **high- or low-value** (`prior_value`) and also carry a continuous **engagement** score. We plant a real, layered moderation: the email lifts low-value customers by **€3** and high-value by **€12 at zero engagement**, **plus** a smooth **+€8 x engagement** slope on top — so at the **average** engagement (0.5) the segment effects work out to **€7** (low) and **€16** (high), which is what the recovery below is graded against. The true per-customer effect is $\tau = 3 + 9 \cdot \text{is_high} + 8 \cdot \text{engagement}$ — a genuine, sizeable, and *continuous* heterogeneity, spanning €3 (low value, zero engagement) to ~€20 (high value, full engagement). `prior_value` also drives spend directly (must be included); there’s a background **trend**.

On real data. There’s no single public dataset for this exact setup, but the swap is trivial: your **own CRM / campaign log** already has a segment column (`prior_value`), covariates, a treatment flag (was the customer emailed) and an outcome (spend). Point the model at those columns instead of the simulator. The one requirement is that the email was assigned randomly *or* that you’ve measured the confounders (notebook 05) — otherwise the “segment effects” are contaminated by who-got-emailed.

The data-generating model — exactly what `dgp.segment_customers` implements (defaults & seed in `src/cmp/dgp.py`). $H = 1[\text{prior_value} = \text{high}]$ with $P(H=1) = 0.4$, engagement $E \sim \text{Beta}(2, 2)$, background trend $G \sim \mathcal{N}(50, 10^2)$, and a **randomized** email $T \sim \text{Bernoulli}(\frac{1}{2})$:

$$\begin{aligned} \tau(H, E) &= 3 + 9H + 8E && \text{(the planted CATE)} \\ \text{spend} &= 5 + \tau(H, E)T + 18H + 4E + 0.3G + \varepsilon, && \varepsilon \sim \mathcal{N}(0, 6^2). \end{aligned}$$

Note the two distinct roles of H : the $18H$ term is prior value’s **direct** effect on spend (high-value customers spend more regardless — why the model must include it), while the $9H$ inside τ is its **moderation** of the email effect — the thing this notebook estimates. At the mean engagement $E = 0.5$ the segment truths quoted above are $3 + 8(0.5) = 7$ € (low) and $3 + 9 + 8(0.5) = 16$ € (high).

TRUE segment-average effects (at mean engagement): `{'low': 7.0, 'high': 16.0}`
 TRUE effect spans €3.0 ... €19.9 across customers

	email	prior_value	engagement	trend	spend	is_high
0	0.0	low	0.918802	58.466377	23.337758	0.0
1	0.0	low	0.651170	60.223782	19.057907	0.0
2	1.0	high	0.242720	53.530129	50.963256	1.0
3	0.0	low	0.119019	37.977148	12.868787	0.0
4	1.0	low	0.488445	61.807028	29.494100	0.0

1.2 3 · Identify — the CATE as a functional of the model

Before fitting anything, we pin down *exactly* what we’re estimating and whether it’s recoverable.

The estimand. The conditional effect, written with the do-operator (intervene, don’t just observe):

$$\tau(x) = \mathbb{E}[\text{spend} \mid \text{do}(\text{email}=1), x] - \mathbb{E}[\text{spend} \mid \text{do}(\text{email}=0), x].$$

Where heterogeneity lives. We assume spend is generated by the structural equation (with e =email, v =is_high, g =engagement)

$$\text{spend} = \beta_0 + \beta_1 e + \beta_2 v + \beta_3 (e \cdot v) + \beta_4 (e \cdot g) + \dots$$

Take the difference between $e=1$ and $e=0$ and everything not multiplied by e cancels, leaving

$$\tau(x) = \beta_1 + \beta_3 v + \beta_4 g.$$

So the email effect is a **baseline β_1 plus moderation terms: all the heterogeneity is carried by the interaction coefficients β_3, β_4** . If those were zero, τ would be the same for everyone and `cate()` would collapse to `ate()`. This is why “is the effect heterogeneous?” is literally the question “is β_3 (or β_4) non-zero?” — which we answer with its posterior in Step 5.

Identification. We still need the email->spend link to be unconfounded. Here email was *randomized*, so unconfoundedness holds by design and the required **adjustment set is empty** — the structural model isn’t doing confounding-correction, it’s giving us a clean way to *query subgroup effects as functionals of the posterior*. `pathmc` reads the graph we declared and confirms that, under that graph, no adjustment is needed — the software checks our drawing, it cannot verify randomization from data:

```
admissible adjustment set(s) for email->spend: [set()]
identifiable: True
```

(An empty adjustment set `[set()]` means “condition on nothing” — correct, because randomization already made email independent of everything. In an *observational* version we’d see the confounders listed here, exactly as in notebook 05.)

1.3 4 • Estimate — fit the structural model

We fit the equation above in one shot; `pathmc` returns a posterior over every coefficient. The `effects_summary()` table below is the direct read-out: each row is a β with its posterior mean and 94% HDI (*Highest Density Interval* — the tightest interval holding 94% of the posterior mass). The two rows to watch are `b_ev` and `b_eg` — the interaction terms. If their intervals sit clearly away from zero, the segments genuinely respond differently; that’s the statistical backing for writing segment-specific rules rather than one blanket policy.

segment model convergence: max r-hat 1.000 - min ESS 3335 - divergences 0

	mean	sd	hdi_3%	hdi_97%
name				
b_e	2.59	0.77	1.18	4.05
b_v	18.39	0.43	17.59	19.21
b_ev	8.13	0.62	6.95	9.26
b_eg	10.19	1.33	7.62	12.58
b_g	3.84	0.93	2.19	5.70
b_tr	0.29	0.01	0.27	0.32

DAG falsification — the check you’d run on real CRM data. Recovery-vs-truth is only available in simulation; on real data you instead test the DAG’s *implied* conditional independences against the data — a wrong DAG shows up as an independence the graph predicts but the data violate.

DAG falsification: 6 implied independences tested, 0 violated at alpha=0.05 (graph consistent with the data).

1.4 5 · Validate — recover the effects, the gap, and the moderation profile

Four checks:

1. **Segment CATEs** vs the planted **€7 / €16** (the low/high segment averages at mean engagement, 0.5) — with credible intervals.
2. **Is the gap real?** the interaction posterior β_3 ; if it clearly excludes zero, the segments truly differ (not noise).
3. **The continuous moderation profile** — CATE as a smooth function of engagement, recovered vs the true $3 + 9 \cdot \text{is_high} + 8 \cdot \text{engagement}$ line. This is the payoff of modelling the continuous moderator rather than crudely binning customers into two groups.
4. **Per-customer recovery** — the effect predicted for each individual customer (from the fitted moderation coefficients) against their true effect, on the 45° line. Recovering *segment averages* is easy; recovering the effect for each *individual* is the real test.

ATE (pooled) €10.80 [10.22060982 11.37241506]

CATE low (eng .5) € 7.69 (true 7.0) [6.98720232 8.45747794]

CATE high (eng .5) €15.82 (true 16.0) [14.88273413 16.71767724]

Interaction beta(email:is_high) mean €8.13 (true 9.0), $P(>0) = > 0.999 \rightarrow$ gap is REAL

Interaction beta(email:engagement) mean €10.19 (true 8.0), 90% credible interval [€7.95, €12.35] — covers the planted €8.0 on this draw; the moderation panel below shows where that slope error pulls the fitted line off the truth



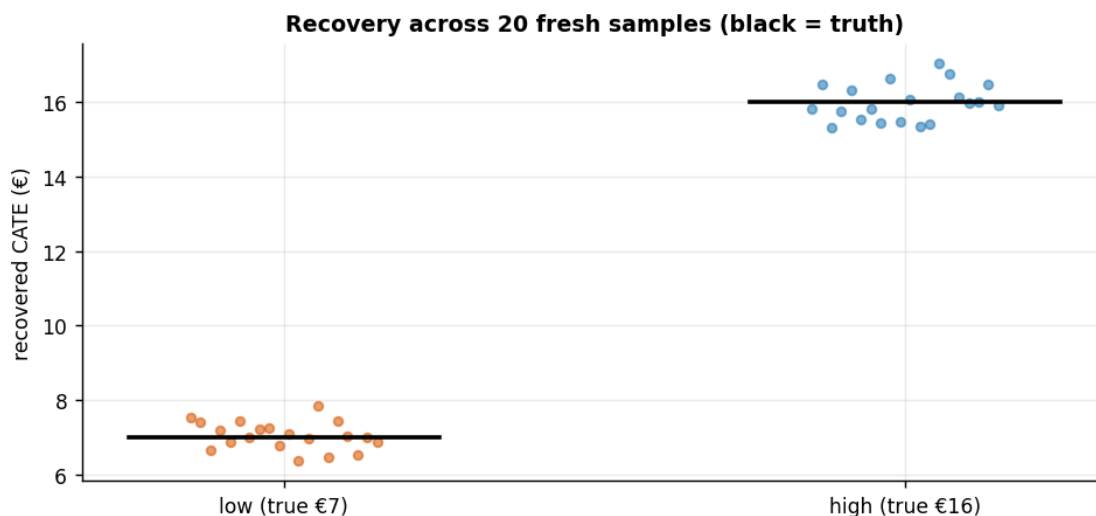
How to read those three panels. *Left* — the two segment-effect posteriors land close to their planted truths (dashed lines) and barely overlap, so the model both **recovers** the effects and **separates** the segments (the low-value posterior runs ~€0.7 high on this draw — §5b shows the bias vanishes across fresh samples). *Middle* — the estimated CATE (solid) tracks the true $3 + 9 \cdot \text{is_high} + 8 \cdot \text{engagement}$ line (dashed) over most of the engagement range — though this draw's interaction slope runs high (€10.2 fitted vs €8.0 planted, printed above), so the fitted line pulls above the truth at high engagement: modelling the *continuous* moderator paid off, because the effect clearly rises with engagement rather than being a flat per-segment number. Note where each line crosses the €8 cost line — that's the **break-even engagement**, the level above which a customer becomes worth emailing; we quantify it (with a 94% HDI) in the decide step below.

Right — predicted vs true effect *per customer* hugs the 45° line (low RMSE), confirming recovery at the individual level, not just on segment averages.

1.4.1 5b · Recovery across many seeds — unbiased *and* calibrated?

A single sample can land a little off; the real test of an estimator is whether it is **centred on the truth over repeated samples** (unbiased, not just lucky here) and whether its intervals **cover** the truth at their stated rate. We refit on many fresh samples and check both. (The per-seed refits are deliberately small, fast fits — enough for the bias/coverage tallies we read off them; their sampler chatter is silenced below.)

low -value: mean €7.05 (true €7) bias +0.05 sd 0.37 · 90% HDI covers truth in 19/20 seeds
 high-value: mean €15.98 (true €16) bias -0.02 sd 0.49 · 90% HDI covers truth in 18/20 seeds



1.5 6 · Decide, in euros — segment rules, and what pooling costs

Now the payoff. First we turn the per-segment effects into concrete **send rules** using the honest probability criterion from notebook 01 — target a segment only where **P(effect > cost) is high** (not merely where the mean beats cost). Then we answer the question that justifies this whole analysis: **how much money does segmenting actually make versus treating everyone as the pooled average?**

We compare four policies on the *known* per-customer truth (only possible in simulation), forming a ladder of increasing sophistication: **pooled** (one all-or-nothing decision on the ATE) -> **segment rules** (one decision per segment) -> **individual** (a per-customer decision on $\hat{\tau}(x)$) -> **oracle** (the unbeatable rule that knows every true effect). The gaps between the rungs are, literally, the euros that finer targeting is worth.

Two rules, two questions. The four-policy ladder below deliberately uses the risk-neutral posterior-mean rule (send whenever *expected* effect beats cost — profit-maximising for expected

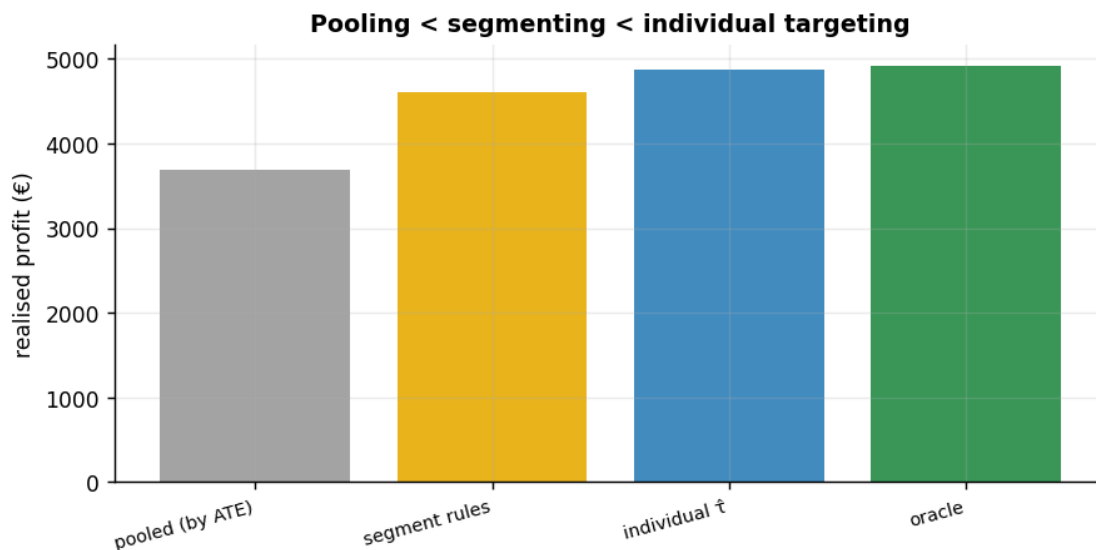
profit), while the per-segment SEND/TEST/SKIP table (printed first in the next cell's output) uses the conservative $P(\text{effect} > \text{cost}) > 0.8$ rule; the two answer different questions — how much money finer targeting is *worth* vs. which segments we are *confident enough* to act on now.

segment	effect	net_of_cost	P(effect>cost)	action
low-value	7.69	-0.31	0.21	SKIP
high-value	15.82	7.82	1.00	SEND

Realised profit by policy:

pooled (by ATE)	€3,682
segment rules	€4,602
individual tau_hat	€4,873
oracle	€4,923

Moving from pooled to individual targeting is worth €1,191 (32% uplift).



Break-even engagement (where CATE = €8), per segment:

low-value : $g^* = 0.53$ (true 0.625) 94% HDI [0.46, 0.61]

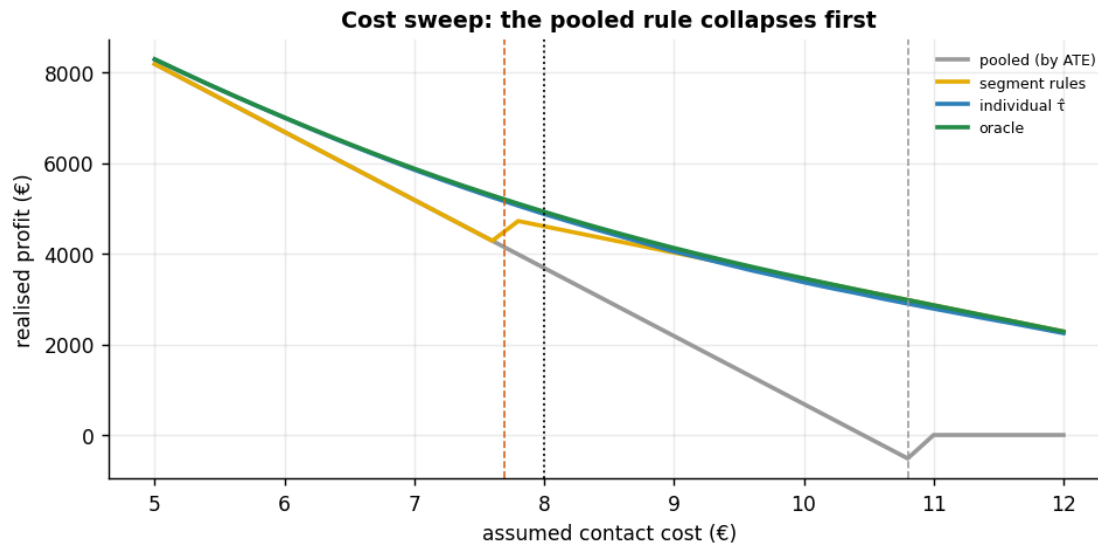
-> SEND to low-value customers only ABOVE engagement ~0.53

high-value: $g^* = -0.28$ (< 0) -> high-value clear €8 at EVERY engagement level (always SEND)

Cost sweep €5-€12: the segment rule drops low-value once cost passes its CATE (€7.7), but the

pooled rule treats EVERYONE up to the ATE (€10.8) then flips to treat-NOBODY - surrendering all the

high-value profit above €10.8, exactly where segment and individual targeting keep earning.



How to read the cost sweep. The four policies earn nearly identical profit only at very low cost, where “email everyone” is optimal. As the contact cost rises they **peel apart at their decision thresholds**: the segment rule stops mailing low-value customers once cost crosses that segment’s CATE (€7.7, orange), while the **pooled rule keeps mailing everyone until cost passes the pooled ATE (€10.8, grey)** — then **collapses to mailing no-one**, forfeiting the high-value profit that individual and segment targeting still capture. That brittleness is the cost of one-size-fits-all: a single averaged decision has no way to keep the profitable segment once the *average* stops clearing cost.

The break-even engagement turns the same €8 logic into an operational cutoff: this sample’s fitted low-value line crosses €8 at engagement **~0.53** (94% HDI [0.46, 0.61]) — *mail a low-value customer only once they are more engaged than that*. The true crossing is 0.625; the fitted cutoff sits a touch lower because this sample’s low-value CATE runs ~€0.7 high (the small solid-vs-dashed gap in the moderation panel), and a break-even — a **ratio** of two estimated coefficients — amplifies that error near the crossing. That is exactly why you carry the interval, not the point, into the decision — with eyes open: in this very sample even the 94% HDI misses the true 0.625 (ratio break-evens inherit *amplified* uncertainty near a crossing), while §5b’s twenty fresh samples show the underlying CATEs are centred. So read the printed band as an honest *starting* range for a cutoff you would confirm on a holdout, not as a guarantee.

1.6 7 · Caveats

- **Heterogeneity is only as good as the moderator you measured.** If the real driver of differential response isn’t `is_high/engagement` (or a proxy), the interaction won’t catch it.
- **Interaction != causing the moderator.** We’re not claiming *making* someone high-value changes responsiveness — only that responsiveness *differs* across pre-existing segments.
- **Random slopes for many thin segments.** With dozens of small segments, replace the single interaction with partially-pooled random slopes so noisy segments borrow strength (see nb 03).

- **Same unconfoundedness caveat** — here email is randomized; observationally a hidden common cause of email and spend would bias every segment effect (see nb 05).